

1. Vector Spaces and Subspaces

A vector space over a field F is a set V together with addition and scalar multiplication satisfying eight axioms: associativity and commutativity of addition, existence of a zero vector, existence of additive inverses, compatibility of scalar multiplication, and the two distributive laws. A subspace is a subset that is itself a vector space under the inherited operations. To verify a subset W is a subspace it suffices to check three conditions: W contains the zero vector, W is closed under addition, and W is closed under scalar multiplication. The intersection of two subspaces is always a subspace, but the union generally is not. The sum of two subspaces is the set of all sums of a vector from each, and is the smallest subspace containing both.

2. Linear Independence and Basis

A finite set of vectors is linearly independent when the only linear combination equal to the zero vector is the one with all coefficients zero. Otherwise the set is linearly dependent, and at least one vector lies in the span of the others. A basis of a vector space is a linearly independent spanning set. Every vector in the space then has a unique representation as a linear combination of basis vectors, and the coefficients in that representation are the coordinates of the vector with respect to that basis. The standard basis of n -dimensional coordinate space consists of the vectors with a single one and zeros elsewhere.

3. Dimension and Rank

All bases of a finite-dimensional vector space have the same number of elements, and that common number is the dimension. The rank of a matrix is the dimension of its column space, which equals the dimension of its row space. The nullity is the dimension of the null space. The rank-nullity theorem states that for a matrix with n columns, rank plus nullity equals n . A matrix has full column rank exactly when its columns are linearly independent, which happens exactly when its null space is trivial. Row operations preserve the row space and the rank but generally change the column space.

4. Linear Transformations

A linear transformation between vector spaces is a map preserving addition and scalar multiplication. Its kernel is the set of vectors mapped to zero, and its image is the set of attained values; both are subspaces. A transformation is injective precisely when its kernel is trivial, and surjective precisely when its image is the whole codomain. Composition of linear transformations is linear. The set of linear transformations from one space to another itself forms a vector space under pointwise operations.

5. Matrix Representation of Transformations

Once bases are fixed for the domain and codomain, every linear transformation between finite-dimensional spaces is represented by a unique matrix whose columns are the images of the domain basis vectors expressed in the codomain basis. Composition of transformations corresponds to matrix multiplication, and invertibility of the transformation corresponds to invertibility of the matrix. The representing matrix depends on the choice of bases, so the same transformation admits many matrix representations.

6. Change of Basis

If two bases are given for the same space, the change-of-basis matrix converts coordinate vectors from one basis to the other, and its columns are the old basis vectors written in the new basis. It is always invertible, with inverse the change-of-basis matrix in the opposite direction. Two matrices A and B represent the same linear operator with respect to different bases exactly when there is an invertible P with B equal to P inverse times A times P . Matrices related in this way are called similar, and similarity preserves the characteristic polynomial, the trace, the determinant and the rank.

7. Determinants

The determinant is the unique alternating multilinear function of the columns that assigns one to the identity. It is zero exactly when the columns are linearly dependent, so a square matrix is invertible precisely when its determinant is nonzero. The determinant of a product is the product of the determinants, and the determinant of a transpose equals that of the original. Expanding along a row or column gives the cofactor expansion. Adding a multiple of one row to another leaves the determinant unchanged, whereas swapping two rows negates it and scaling a row scales it.

8. Eigenvalues and the Characteristic Polynomial

A scalar λ is an eigenvalue of a square matrix A when $A - \lambda I$ is singular, equivalently when the characteristic polynomial, defined as the determinant of $A - \lambda I$, vanishes at λ . The characteristic polynomial of an n by n matrix has degree n , so over the complex numbers there are n eigenvalues counted with algebraic multiplicity. The sum of the eigenvalues equals the trace and their product equals the determinant. Similar matrices have identical characteristic polynomials and therefore identical eigenvalues.

9. Eigenvectors and Eigenspaces

An eigenvector for the eigenvalue λ is a nonzero vector v with A times v equal to λ times v . The eigenspace of λ is the null space of $A - \lambda I$, and its dimension is the geometric multiplicity of λ . The geometric multiplicity never exceeds the algebraic multiplicity. Eigenvectors corresponding to distinct eigenvalues are linearly independent, so a matrix with n distinct eigenvalues has a basis of eigenvectors. An eigenvalue whose geometric multiplicity is strictly less than its algebraic multiplicity is called defective.

10. Diagonalization

A square matrix is diagonalizable when it is similar to a diagonal matrix, equivalently when it has a basis of eigenvectors. In that case A equals P times D times P inverse, where the columns of P are eigenvectors and the diagonal entries of D are the corresponding eigenvalues. A matrix is diagonalizable exactly when for every eigenvalue the geometric multiplicity equals the algebraic multiplicity, so a defective matrix is not diagonalizable. Diagonalization makes powers cheap, since A to the k equals P times D to the k times P inverse, and D to the k just raises the diagonal entries.

11. Symmetric Matrices and Orthogonal Diagonalization

A real matrix is symmetric when it equals its own transpose. The spectral theorem states that a real matrix is orthogonally diagonalizable precisely when it is symmetric. Every real symmetric matrix therefore has real eigenvalues and admits an orthonormal basis of eigenvectors, so it can be written as Q times D times Q transpose with Q orthogonal. Eigenvectors belonging to distinct eigenvalues of a symmetric matrix are automatically orthogonal, which is why the orthonormal basis exists without further hypotheses.

12. Inner Product Spaces

An inner product assigns a scalar to each pair of vectors, linearly in one argument, conjugate-symmetric, and positive definite on nonzero vectors. It induces a norm as the square root of the inner product of a vector with itself. The Cauchy-Schwarz inequality bounds the absolute inner product by the product of the norms, and the triangle inequality follows from it. Two vectors are orthogonal when their inner product vanishes, and the Pythagorean identity holds for orthogonal pairs. The orthogonal complement of a subspace is the set of vectors orthogonal to all of it.

13. Gram-Schmidt Orthogonalization

The Gram-Schmidt process converts a linearly independent list into an orthogonal one spanning the same subspaces at every stage. Each new vector has subtracted from it its orthogonal projections onto all previously produced vectors, and the result is then normalized to obtain an orthonormal sequence. The process fails exactly when the input is linearly dependent, because some vector would be reduced to zero. Applied in floating point the classical form loses orthogonality, so the modified variant, which subtracts projections one at a time against the updated vector, is preferred numerically.

14. QR Decomposition

Every matrix with linearly independent columns factors as Q times R , where Q has orthonormal columns and R is upper triangular with positive diagonal entries. Applying Gram-Schmidt to the columns produces exactly this factorization, with R recording the projection coefficients. In practice Householder reflections or Givens rotations are used instead, being numerically stable. The factorization solves least squares problems without forming the normal equations, which is important because forming them squares the condition number.

15. Singular Value Decomposition

Every matrix, square or not, factors as U times Σ times V transpose, with U and V orthogonal and Σ diagonal with nonnegative entries called the singular values, conventionally in decreasing order. The singular values are the square roots of the eigenvalues of A transpose times A . The number of nonzero singular values equals the rank. Truncating the decomposition after k terms gives the best rank- k approximation in both the spectral and Frobenius norms, which is the Eckart-Young theorem. The condition number is the ratio of largest to smallest singular value.

16. Least Squares Problems

When a linear system has no solution, the least squares problem seeks the vector minimizing the Euclidean norm of the residual. The minimizer satisfies the normal equations, obtained by multiplying through by the transpose, and is unique exactly when the matrix has full column rank. Geometrically the fitted value is the orthogonal projection of the target onto the column space, so the residual is orthogonal to every column. When the matrix is rank deficient the minimum-norm solution is singled out using the pseudoinverse.

17. Positive Definite Matrices

A symmetric matrix is positive definite when the quadratic form is strictly positive for every nonzero vector, and positive semidefinite when it is merely nonnegative. Equivalent characterizations are that all eigenvalues are positive, that all leading principal minors are positive, and that a Cholesky factorization into a lower triangular factor times its transpose exists. Positive definite matrices are invertible and their inverses are positive definite. The Cholesky factorization costs about half a general LU factorization, which is why it is preferred whenever definiteness is known.

18. Jordan Canonical Form

Over the complex numbers every square matrix is similar to a Jordan form: a block diagonal matrix whose blocks have a single eigenvalue on the diagonal and ones on the superdiagonal. The number of blocks for an eigenvalue equals its geometric multiplicity and the total size of those blocks equals its algebraic multiplicity, so the form is diagonal exactly when the matrix is diagonalizable. The form is unique up to ordering of the blocks. It shows that the minimal polynomial is the product over eigenvalues of the factor raised to the largest block size.